

INFOGRAPHIC: Bootstrapping & Jackknifing

Resampling Methods in Statistical Modelling

[BOOTSTRAPPING]

Resampling **with replacement**. Generates thousands of new synthetic samples of size n from the original data.

[JACKKNIFING]

Resampling **without replacement**. Systematically leaves out exactly one observation (or a block) at a time.

1. The Problem They Solve

Classical statistical formulas for estimating standard errors, bias, or confidence intervals depend heavily on strict assumptions (like normality). When an estimator is highly complex (e.g., medians, ratios, or non-linear coefficients), analytical equations often do not exist. Resampling methods use computational power to let the data speak for itself.

2. Bootstrapping: The Computational Powerhouse

Proposed by Bradley Efron in 1979, Bootstrapping treats your original sample as if it were the entire population.

The Bootstrap Process

1. Take your original dataset of size n .
2. Draw a new sample of size n **with replacement** (meaning some data points will repeat, and others will be left out).
3. Calculate your statistic of interest (e.g., mean, median, β -coefficient) on this new bootstrap sample.
4. Repeat this loop B times (typically $B = 1000$ or $B = 10000$).

The resulting distribution of those B statistics allows you to directly measure standard errors and construct **Bootstrap Confidence Intervals** (such as Percentile or BCa intervals).

3. Jackknifing: The Systematic Approach

Invented earlier by Maurice Quenouille (1949), Jackknifing is a deterministic, non-random resampling framework.

The Jackknife Process ("Leave-One-Out")

If you have a dataset of size n , you create exactly n new sub-samples. In each sub-sample i , you delete exactly the i -th observation. The size of each sub-sample is $n - 1$.

Pseudo-values Calculation:

For each omitted sample, we calculate a "pseudo-value" J_i that reflects the impact of that specific data point on our target statistic.

$$J_i = n \theta - (n - 1) \theta_{(-i)}$$

Where θ is the full-sample estimate and $\theta_{(-i)}$ is the estimate with the i -th point removed.

4. Head-to-Head Comparison

Feature	Bootstrapping	Jackknifing
Replacement	With replacement	Without replacement
Nature	Stochastic / Random simulations	Deterministic / Exact sub-samples
Number of Samples	Chosen by user (e.g., $B = 2000$)	Fixed by data size (n samples)
Computational Load	High (demands thousands of iterations)	Low (only n loops required)
Failure Points	Extremely small sample sizes	Non-smooth statistics (e.g., Medians)

5. Critical Limitations in Statistical Modelling

- **The Smoothness Constraint:** Jackknifing fails completely for "non-smooth" parameters like the median or mode, because deleting one point might change the median drastically or not at all, creating unstable variance estimates.
- **Dependent Data Warning:** Standard versions of both methods assume observations are independent (*i.i.d.*). For time-series, spatial, or clustered data, you must use specialized frameworks like **Block Bootstrapping** or **Block Jackknifing** to protect the internal data correlations.